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Unified chromatography in drug development: Exploiting chaotropic/kosmotropic salts for an accelerated method development

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A B S T R A C T

Recent trends in supercritical fluid chromatography (SFC) introduced an innovative gradient profile called Unified Chromatography (UC), which pushes the amount of liquid modifier up to 80–100 % of the total mobile phase composition. These new conditions allow the full transition from a supercritical to a liquid state, unifying the benefits of both SFC and liquid chromatography. However, to facilitate the use of UC for industrial drug development, a stronger effort is needed to streamline and simplify its method development and optimization. In this work, a quick and novel method development procedure for UC is introduced, enabled by the first-time use of novel additives in SFC/UC that exploit chaotropic/kosmotropic properties. A comprehensive view on some fundamental properties, such as the amount of liquid modifier blended with supercritical CO_2 ($scCO_2$) and the percentage of water added in the mobile phase is given, to clarify the benefits of using either a chaotropic salt ($NaClO_4$), kosmotropic (HCOONa) or salt with mixed properties (NaOMs – sodium methanesulfonate). With this expanded knowledge, challenging separations of nucleosides, nucleotide, indoles, triazoles and related derivates have been accomplished with UC. Finally, we provide an example of UC delivering a faster and better method for an AbbVie pipeline compound under accelerated stability study. The combined use of $scCO_2$-based chromatography and the novel additive $NaClO_4$ ensures the retention and elution of all degradation species generated at different conditions, where RP-HPLC failed to provide satisfactory performance.

1. Introduction

Recent trends seen in the pharmaceutical industry indicate a tremendous increase in new and challenging chemical entities being developed at early and late stages of the drug development process [1,2]. The increased number is tied also to a higher degree of complexity of these new molecules [3]. This translates into a stronger need to focus on either optimizing the already existing analytical methodologies, or to propose new and better ones in the field of drug discovery [4–6].

Chromatography occupies, among all techniques available in the analyst toolbox, a predominant role [4]. Ultra-high-performance liquid chromatography (UHPLC) and gas chromatography (GC) have largely proven themselves to be the much-needed workhorse for analytical laboratories. Recently, new approaches have been devised, such as multi-dimensional liquid chromatography (mD-LC) to tackle the new chemical and biochemical entities under development by pharmaceutical industries [7–9]. One of the main drawbacks of mD-LC lies in its inherent complexity and lengthy method development, which has limited its widespread use in academic and industrial settings. Although some works have successfully introduced methodologies to overcome these constraints, there is still much to be done to efficiently integrate mD-LC in the daily routine of analytical scientists [10,11].

Alternative approaches to liquid chromatography have also been explored. Among these, supercritical fluid chromatography (SFC) is certainly considered as one of the most successful techniques [4,12–16]. SFC utilizes a super/subcritical mobile phase by blending apolar supercritical carbon dioxide ($scCO_2$), with a polar organic liquid modifier (e.g. methanol – MeOH, ethanol – EtOH, etc.) to ensure the elution of the more retained samples. One key peculiarity of SFC is the ability to retain a much higher diversity of (bio)chemical compounds when compared to LC techniques such as RPLC (poorly suited for highly polar low molecular weight species) or HILIC (poor retention of hydrophobic molecules) [17–19]. With the introduction of ultra-high performance supercritical fluid chromatography (UHPSFC) instrumentation in the early 2010s, $scCO_2$-based separations have been increasingly developed in various environments. Further trends such as the systematic addition of water (in various amounts) blended with methanol and $scCO_2$ have enabled SFC to be better suited for highly polar molecules, expanding its range

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[19–24].

A recent tendency seen with SFC is the use of unorthodox gradient profiles, pushing the amount of liquid modifier in the mobile phase above the canonical 40–50 % to reach extremely high values, 70–80 % or even 100 % thus obtaining a fully liquid mobile phase. These conditions enabled the transition from SFC to unified chromatography (UC) [25,26]. The rationale behind UC is to further enhance the elution strength of the mobile phase, aiming at improving the compatibility of SFC with highly polar biochemical species with either low or relatively high molecular weight (e.g. protein, peptides) while maintaining the excellent performance normally delivered for apolar or slightly polar small molecules. As new concepts further optimizing the original idea behind UC are being developed, UC is now mature enough to be introduced in industrial settings, to become a viable alternative to analysts for the separation of highly complex multicomponent mixtures [19]. However, similarly to SFC, the method development process for UC is seen as very challenging, requiring both column and mobile phase screenings. This is generally viewed as tedious and time-consuming, thus representing a key constraint for further UC use in unexplored areas.

In this work, we propose an easy and quick procedure for efficient method development of UC methods within industrial drug development settings, by exploiting a series of new and unorthodox salts (sodium perchlorate – $NaClO_4$; sodium methanesulfonate – NaOMs; sodium formate – HCOONa) possessing orthogonal chaotropic/kosmotropic properties. An investigation of fundamental properties, such as the column chemistry influence (acidic/basic stationary phases) and the mobile phase composition (amount of water mixed with methanol and $scCO_2$) is presented, using a series of modified nucleosides and nucleotides eluting across the entire UC gradient range herein considered. Final optimized methods for different mixtures of pharmaceutically relevant compounds are provided, illustrating the easiness and power of this facilitated UC method development approach. Finally, an example of a proprietary compound developed *in-house* is given, showing the advantage of UC over the more conventional HPLC for drug development examples.

2. Materials and methods

2.1. Chemical, reagents and sample preparation

Supercritical carbon dioxide (scCO2) of food grade (3.0) was purchased from Airgas (Radnor, Pennsylvania, USA). Acetonitrile (ACN) and methanol (MeOH) of HPLC grade were purchased from Sigma-Aldrich (St. Louis, MO, USA), while Milli-Q grade water was obtained from a Milli-Q Type 1 Ultrapure System by Millipore Sigma (Burlington, MA, USA). Sodium perchlorate ($NaClO_4$), sodium methanesulfonate (NaOMs), sodium formate (NaForm – HCOONa) and ammonium acetate were purchased from Sigma-Aldrich. Mixture 1 (6-dimethylaminopurine; 2-chloro-N6-cyclopentyladenosine; 6-methylmercaptopurine riboside; 5-fluorouridine; 2-amino-6-methylmercaptopurine; azathioprine; 6-azauridine; 4-thiouridine; 7-methylguanosine; thymidine 3,5-cyclic monophosphate) and mixture 2 (1-(methoxymethyl)-1*H*-benzotriazole; *N*-(4-formylpentyl)carbazole; 6-methoxy-7-azaindole; 3-[5-(4-chlorophenyl)furan-2-yl]acrylic acid; 4-methyl-1*H*-benzotriazole; 5-methyl-1*H*-benzotriazole; 1*H*-benzotriazole-1-methanol; benzotriazole; 7-azaindole-carboxyaldehyde; benzotriazole-5-carboxylic acid) components were all purchased from Sigma-Aldrich. All samples were dissolved in either ACN:water 50:50 (Mixture 1) or pure ACN (mixture 2) at an initial concentration of 2.0 mg mL^{-1}, then subsequently diluted with pure ACN to a final concentration of 0.4 mg mL^{-1} (mixture 1) and 0.5 mg mL^{-1} (mixture 2). For all AbbVie-1 samples (control and accelerated stability), methanol was used as solvent and a final concentration of 1.0 mg mL^{-1} was used.

2.2. Instrumentation, columns and screening conditions

Two columns were chosen for this study: Torus DEA (diethylamino) and Torus DIOL from Waters (Milford, MA, USA), all with the following column dimensions: 100 × 3.0 mm I.D. Both columns were packed with 1.7 µm fully porous silica particles.

All SFC/UC analyses were performed on a Waters Acquity UPC^2 system, equipped with a binary solvent manager (BSM) pump, an autosampler fitted with a 10 µL injection loop, a multicolumn oven with active pre-heaters, capable to host up to eight columns, a PDA detector with a 8.4 µL flow-cell and an active+passive backpressure regulator (ABPR) module. For RP-HPLC runs, an Agilent 1260 was used, composed of a binary pump, an autosampler fitted with a 50 µL injection loop, a column oven and a UV detector. Instrument control for both SFC/UC and RP-HPLC analyses was performed with Empower v3.0 (Waters), while data treatment was performed with Microsoft Excel (Redmond, WA, USA).

For the generic UC screening conditions, the following parameters were as follows: a gradient ramp from 5 % to 80 % in 7 min, hold-up at 20:80 $scCO_2$:B for 1 min, then return to initial conditions in 0.1 min. Total run time: 10 min. Flow-rate: 0.75 mL min^{-1}. Column temperature: 40 °C. ABPR: 110 bar. Injection volume 1.0 µL.

For the experiments described in Section 3.4, the following experimental conditions are as follows: UC separations were achieved on a DIOL column, using 95:5 MeOH:H_{2O} + 5 mM $NaClO_4$ as the co-solvent mixed with $scCO_2$. Flow-rate: 1.00 mL min^{-1}. Column temperature: 45 °C. ABPR: 110 bar. Detection: UV @ 268 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 3.0 µL. Gradient profile: initial hold at 95:5 $scCO_2$:B for 0.5 min, first ramp to 40 % B in 4.5 min, followed by a second ramp to 70 % B in 4.0 min, isocratic hold at 30:70 $scCO_2$:B for 1.0 min then return to initial conditions in 0.1 min for a total run time of 11.0 min. RP-HPLC conditions were as follows: column used was a Waters X-Bridge C18 (150 × 4.6 mm I.D. – 3.5 µm). Mobile phase A was 10 mM ammonium acetate in water, while mobile phase B was acetonitrile pure. Gradient profile: isocratic hold at 95:5 A:B for 3.0 min, increase to 40 % B in 7.0 min, then 45 % B in 6.0 min, finally to 90 % B in 5.0 min, isocratic hold at 10:90 A:B for 3.0 min, then return to initial conditions in 0.1 min for a total run time of 30.0 min. Column temperature was 40 °C. Flow-rate: 1.0 mL min^{-1}. Detection: UV @ 268 nm. Injection volume: 10.0 µL. Injection solvent: pure methanol.

3. Results and discussion

3.1. Use of chaotropic/kosmotropic agents in $scCO_2$-based separations

The Hofmeister series has been traditionally used to classify cations and, more importantly, anions based on their abilities to promote (or not) protein solubility in water. Chaotropes and kosmotropes can achieve so due to their peculiar interaction with water molecules: chaotropes break down the structuring of water, thus helping protein solubility in it. On the contrary, kosmotropes promote the water structuring, which in consequence contributes to the "salting-out" of proteins and the increase of protein hydrophobicity [27]. The use of salts based on their chaotropic/kosmotropic properties (following the Hoffmeister series) is not new in chromatography. Traditionally, chaotropic agents such as perchlorate anion (ClO_4^-), hexafluorophosphate (PF_6^-), and trifluoroacetate (CF_3COO^-) have been used in RPLC to obtain overall better chromatographic performance (peak shape, increased retention of early-eluting basic compounds, etc.) via the combination of different effects induced by the chaotropic salt [28–30]. Kosmotropic salts, on the other hand, are regularly used in hydrophobic interaction chromatography (HIC) to exploit their "salting-out" properties, reducing the overall solvation of proteins and thus increasingly expose their hydrophobic regions. This ensures a better retention onto the HIC stationary phase [31–33]. With $scCO_2$-based separations, however, the scenario is completely different. Only one report describes the systematic use of

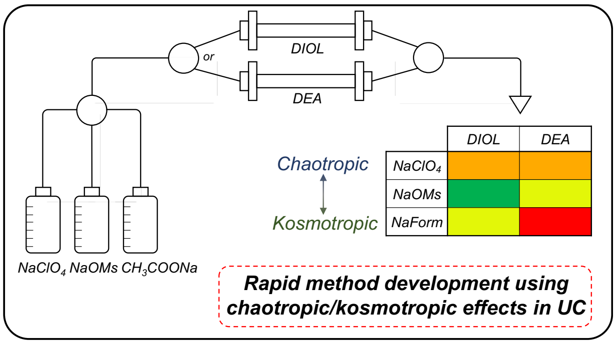


Fig. 1. Schematics of the method development approach introduced for UC.

chaotropic effects in SFC [34], by exploiting the in situ formation of the carbonate ion generated by the addition of ammonia (NH_4OH) and water in the mobile phase. This work illustrates the clear advantages of introducing chaotropic effects with SFC, enabling a much-improved separation of highly polar chemical species that have been historically challenging with this technique. In the same work, the authors clarified that orthodox chaotropic agents, such as the one previously mentioned, cannot be used in SFC due to their poor solubility in the super/-subcritical mobile phase. While this is true for SFC, the use of UC conditions may improve the scenario and allow an expanded use of chaotropic/kosmotropic effects also with $scCO_2$-based chromatography, with the ultimate goal of introducing a simple and effective UC method development strategy to be easily deployed within pharmaceutical industrial settings.

Utilizing salts with either salting-in or salting-out properties can be beneficial to streamline and simplify method development for any chromatographic technique [29]. Thus, a new method development approach is herein introduced for unified chromatography, to expand its range and user-friendliness (Fig. 1). Three different salts have been chosen for this UC screening strategy: sodium perchlorate ($NaClO_4$), sodium methanesulfonate (NaOMs) and sodium formate (HCOONa). The selection was dictated so to have a chaotropic agent (sodium perchlorate), a kosmotropic one (sodium formate) and a salt with intermediate properties (sodium methanesulfonate) [35–37]. While the use of formate salts is common in SFC and UC, to the best of the authors knowledge this is the first use of $NaClO_4$ and NaOMs for $scCO_2$-based chromatography. Their addition is only possible thanks to the uniqueness of unified chromatography; as the liquid modifier reaches elevated amounts in the mobile phase (higher than 50 %, up to 100 %), it greatly reduces the salt precipitation issue previously reported [34]. On the other hand, the use of sodium-based salts impedes the possibility to hyphenate UC with mass spectrometers (MS), as already seen with the use of chaotropic/kosmotropic salts in liquid chromatography.

Column selection is also key for SFC/UC: to streamline the entire method development process, the choice has been limited to two UHPSFC stationary phases (Waters Torus DEA and DIOL) to simply the column selection process. The two stationary phases present complementary properties among them (DEA – basic selector; DIOL – acidic/neutral selector) which make them well-suited for acidic and neutral analytes (Torus DIOL) or basic compounds (Torus DEA). These columns are also capable to deliver excellent performance for $scCO_2$-based separations with a high degree of robustness, thanks to their reduced number of free silanols which limits the formation of methyl-silyl ethers on the silica particle surface [38] and the use of long and polar column selectors which help shielding the free silanols. Overall, the entire method screening takes less than two hours per compound, delivering a

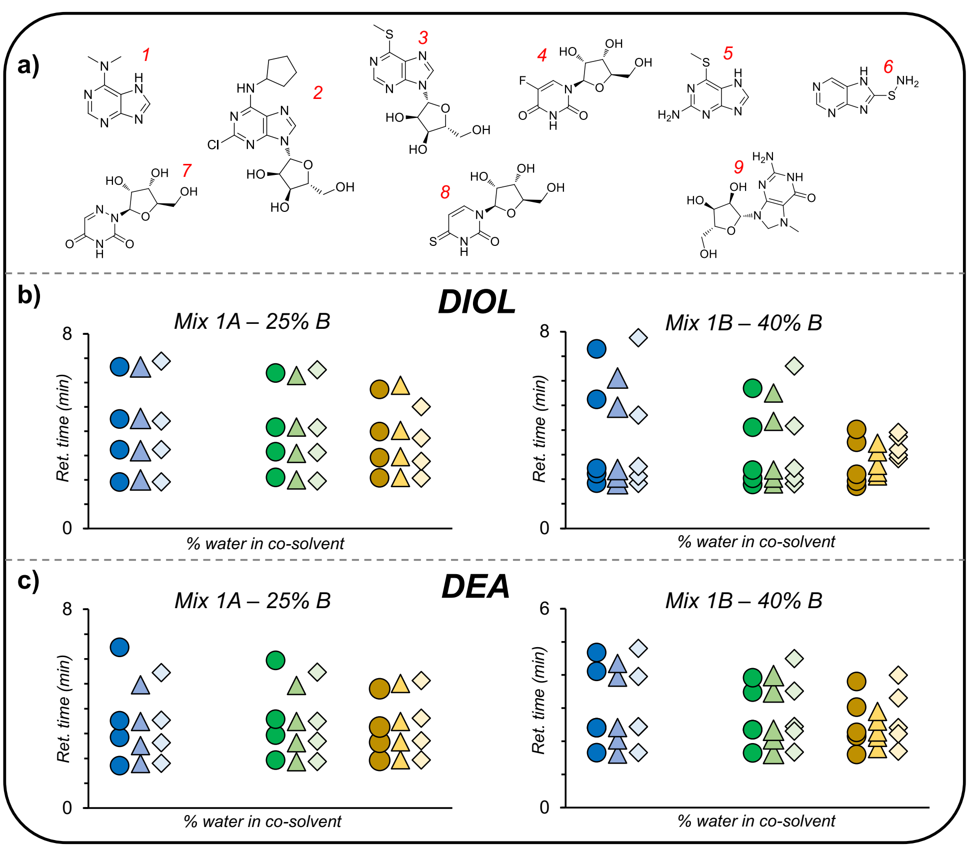


Fig. 2. (a) Structures of the model compounds used for mixture 1A (**1** - 6-dimethylaminopurine; **2** - 2-chloro-N6-cyclopentyladenosine; **3** - 6-methylmercaptopurine riboside; **4** - 5-fluorouridine) and 1B (**5** - 2-amino-6-methylmercaptopurine; **6** - azathioprine; **7** - 6-azauridine; **8** - 4-thiouridine; **9** - 7-methylguanosine), ordered according to their retention times (b) Retention time vs.% water in co-solvent plots obtained from isocratic runs performed with 75:25 $scCO_2$:B (left) for mixture 1A, 60:40 $scCO_2$:B (right) for mixture 1B on a Torus DIOL column. Flow-rate: 0.75 mL min^{-1}. Column temperature: 40 °C. ABPR: 110 bar. Detection: UV @ 254 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 1.0 μL. Plots legend: blue – 98:2 MeOH:H_2O; green – 95:5 MeOH:H_2O; gold – 93:7 MeOH:H_2O. Circle symbols: 5 mM NaClO4 in co-solvent. Triangle symbols: 5 mM NaOMs in co-solvent. Diamond symbols: 5 mM NaForm. (c) Retention time vs.% water in co-solvent plots obtained from isocratic runs performed with 75:25 $scCO_2$:B (left) for mixture 1A, 60:40 $scCO_2$:B (right) for mixture 1B on a Torus DEA column. Flow-rate: 0.75 mL min^{-1}. Column temperature: 40 °C. ABPR: 110 bar. Detection: UV @ 254 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 1.0 μL. Plots legend: blue – 98:2 MeOH:H_2O; green – 95:5 MeOH:H_2O; gold – 93:7 MeOH:H_2O. Circle symbols: 5 mM NaClO4 in co-solvent. Triangle symbols: 5 mM NaOMs in co-solvent. Diamond symbols: 5 mM NaForm.(For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Resolution (*Rs*) values for mixture 1A isocratic separations illustrated in Fig. 2 (b) and (c). Peaks 1–2: 6-dimethylaminopurine and 2-chloro-N6-cyclopentyladenosine. Peaks 2–3: 2-chloro-N6-cyclopentyladenosine and 6-methylmercaptopurine riboside. Peaks 3–4: 6-methylmercaptopurine riboside and 5-fluorouridine.

Column	MeOH:H_2O ratio	Additive	Resolution Peaks 1–2	Resolution Peaks 2–3	Resolution Peaks 3–4
DIOL	98:2	$NaClO_4$	11.15	7.99	11.29
		NaOMs	10.85	6.88	11.02
		NaForm	11.19	6.95	12.08
	95:5	$NaClO_4$	8.78	6.85	12.50
		NaOMs	9.29	7.27	11.71
		NaForm	10.27	7.01	13.17
	93:7	$NaClO_4$	5.86	6.14	9.48
		NaOMs	5.66	6.23	9.94
		NaForm	5.04	5.39	6.24
DEA	98:2	$NaClO_4$	9.03	4.44	15.09
		NaOMs	5.94	6.76	8.44
		NaForm	6.92	6.27	10.71
	95:5	$NaClO_4$	8.00	4.33	12.37
		NaOMs	7.05	6.03	6.24
		NaForm	6.24	5.79	11.40
	93:7	$NaClO_4$	4.95	3.70	8.18
		NaOMs	5.07	4.98	8.61
		NaForm	5.19	5.08	7.98

complete landscape of what combination of chaotropic/kosmotropic agent + column chemistry works best for a given mixture. This methodology has the potential to greatly expand the usability of UC within an area (drug development) where unified chromatography has been rarely used.

3.2. Optimization of the screening conditions

Once the initial set-up has been chosen, the following step is to understand the effect of mobile phase composition. The use of UC-based gradients allows higher amounts of water mixed with methanol [19,39,40], which remains the preferred choice as the organic modifier. Therefore, the proportions of methanol and water pre-blended in the liquid modifier have been chosen as the next focusing point. Although water is generally considered as an additive for SFC/UC separations to improve the overall chromatographic performance, some of the latest works have begun utilizing water in higher amounts [19,40]. As chaotropic/kosmotropic salts impact, among other things, hydrogen bonds, it becomes important to understand whether different ratios of MeOH:H_2O may generate different chromatographic profiles with the presence of either $NaClO_4$, NaOMs or NaForm.

To do so, a series of modified nucleosides and nucleobases (Fig. 2(a)) has been selected to carry out fundamental studies on how each of the three salts herein considered ($NaClO_4$ – NaOMs – NaForm) behaves at different percentages of water (2–5–7 %) blended with methanol and, ultimately, $scCO_2$. To avoid potential interferences from the mobile phase compression phenomenon, isocratic runs were performed instead of gradient ones. Retention times observed under 75:25 and 60:40 $scCO_2$:B mobile phases were recorded for both mixtures on the DIOL (Fig. 2(b)) and DEA columns (Fig. 2(c)). Experimental conditions were fixed so to have retention and elution of all model compounds in the range of 1 to 8 min (retention factor range of ~2–15). The concentration of each additive was fixed at 5 mM, to avoid potential precipitation issues at higher amounts (>10 mM) while exploiting, at the same time, the chaotropic/kosmotropic effects.

From Fig. 2(b), (c) and Tables 1, 2 some trends emerge. Among all conditions tested, none of the salts (sodium perchlorate, methanesulfonate and formate) always provides the best separation quality on both columns. Specifically, the use of the chaotropic salt ($NaClO_4$) was the worst one for mixture 1B separation on the DEA column at both 2 % and 5 % of water, providing separation of only 4 out of the 5 molecules present in mixture 1B. At 7 % of water in methanol, the overall separation shows improved separation performance (Fig. 2(c)) for $NaClO_4$, as illustrated also by the resolution (Rs) values given in Table 2. The use of the kosmotropic salt (sodium formate – HCOONa) is the clear option (93:7 MeOH:H_2O co-solvent), followed by the milder kosmotropic additive sodium methanesulfonate (98:2 and 95:5 MeOH:H_2O co-solvents) (Table 2). This may be due to the better shielding of the stationary phase positive charge from the water present in the mobile phase when kosmotropic salts are used, thanks to their water-structuring enhancing effect. For the same mixture 1B, $NaClO_4$ provides consistently better results on the non-ionizable stationary phase (DIOL column) under all amounts of water (2–5–7 %) (Table 2). Isocratic runs performed with the 93:7 MeOH:water co-solvent for mixture 1B on this column show how the chaotropic additive was the only additive that enabled a partial separation of all 5 components, while sodium methanesulfonate and sodium formate failed to do so (Fig. 2(c) and Table 2). Mixture 1A, on the other hand, was easily separated under all combinations of MeOH:H_2O co-solvents and additives herein tested (Fig. 2(b)). Minor differences are visible (Table 1); however, these may be due to specific interactions between the salt anion employed and the analytes.

Once the effect of different water percentages on the model compounds separation have been investigated with each chaotropic/

Table 2
Resolution (*Rs*) values for mixture 1B isocratic separations illustrated in Fig. 2(b) and (c). Peaks 5–6: 2-amino-6-methylmercaptopurine and azathioprine. Peaks 6–7: azathioprine and 6-azauridine. Peaks 7–8: 6-azauridine and 4-thiouridine. Peaks 8–9: 4-thiouridine and 7-methylguanosine.

Column	MeOH:H_2O ratio	Additive	Resolution Peaks 5–6	Resolution Peaks 6–7	Resolution Peaks 7–8	Resolution Peaks 8–9
DIOL	98:2	$NaClO_4$	2.96	1.74	19.19	9.14
		NaOMs	2.11	2.34	17.56	5.80
		NaForm	2.10	2.89	14.21	8.64
	95:5	$NaClO_4$	1.88	2.59	13.42	8.61
		NaOMs	1.74	2.58	15.13	5.86
		NaForm	1.89	3.26	13.30	9.07
	93:7	$NaClO_4$	1.15	1.63	8.13	2.39
		NaOMs	0.58	1.30	3.41	1.94
		NaForm	0.35	0.98	1.48	0.89
DEA	98:2	$NaClO_4$	5.85	N/A	7.75	4.19
		NaOMs	3.67	3.06	9.88	3.73
		NaForm	6.38	N/A	10.35	5.16
	95:5	$NaClO_4$	5.73	N/A	6.39	2.79
		NaOMs	3.80	2.30	9.61	3.24
		NaForm	5.11	1.10	8.79	5.22
	93:7	$NaClO_4$	3.28	1.84	5.60	3.20
		NaOMs	1.14	1.92	1.11	3.07
		NaForm	3.35	1.47	6.87	2.77

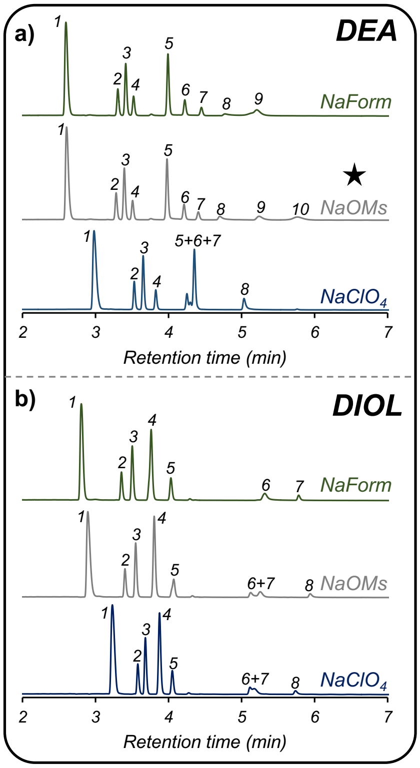


Fig. 3. (a) Chromatograms obtained for mixture 1 under gradient mode on a Torus DEA column using 5 mM sodium formate (green), 5 mM sodium methanesulfonate (grey) or 5 mM sodium perchlorate (blue) in 95:5 MeOH:H_2O co-solvent. Gradient profile: 5 % to 80 % in 7 min, hold-up at 20:80 $scCO_2$:B for 1 min, then return to initial conditions in 0.1 min. Total run time: 10 min. Flow-rate: 0.75 mL min^{-1}. Column temperature: 40 °C. ABPR: 110 bar. Detection: UV @ 254 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 1.0 μL. (b) Chromatograms obtained for mixture 1 under gradient mode on a Torus DIOL column using 5 mM sodium formate (green), 5 mM sodium methanesulfonate (grey) or 5 mM sodium perchlorate (blue). Gradient profile: 5 % to 80 % in 7 min, hold-up at 20:80 $scCO_2$:B for 1 min, then return to initial conditions in 0.1 min. Total run time: 10 min. Flow-rate: 0.75 mL min^{-1}. Column temperature: 40 °C. ABPR: 110 bar. Detection: UV @ 254 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 1.0 μL.(For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

kosmotropic additive, a final choice for the co-solvent composition can be reached for the method development strategy herein proposed. The use of 93:7 MeOH:H_2O co-solvent clearly benefits from the use of kosmotropic sodium formate with an ionizable stationary phase (Fig. 2(c)), especially for those analytes eluting at higher percentages of co-solvent in the mobile phase. On the other hand, the chaotropic sodium perchlorate is the favorite choice with a column less prone to ionization (Fig. 2(b)). The 93:7 MeOH:H_2O co-solvent composition can be, therefore, used for highly retained compounds, however it may be poorly suited for early-eluting apolar analytes. On the other hand, the use of 98:2 MeOH:H_2O modifier may provide precipitation issues of the additives here considered, in particular of sodium perchlorate. Finally, the choice fell on the use of 95:5 MeOH:H_2O as the best compromise.

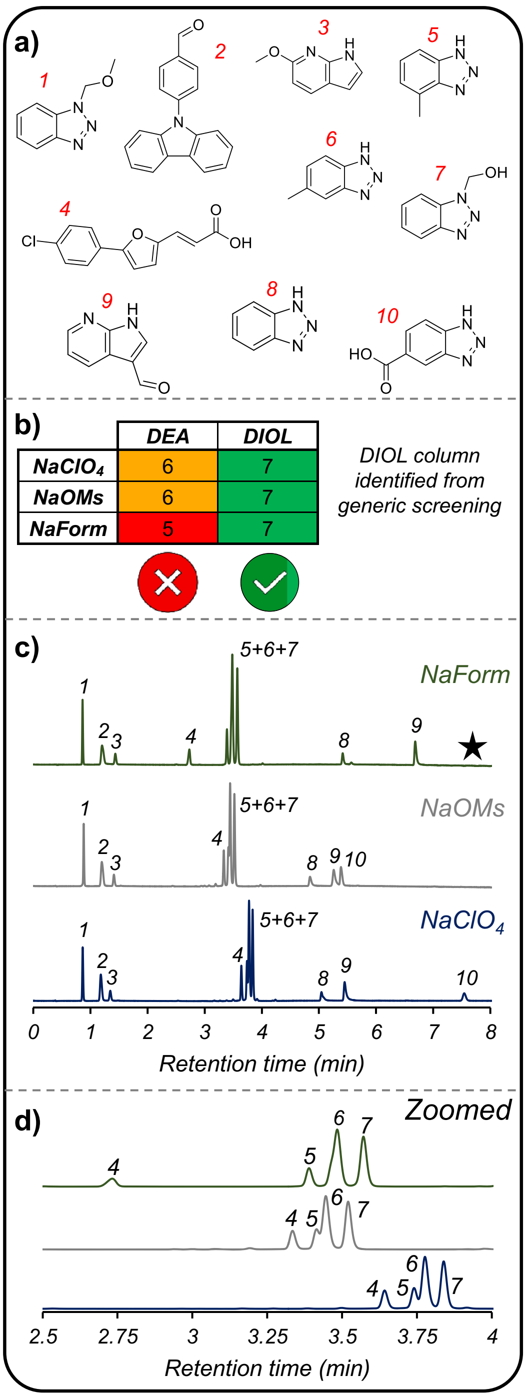
3.3. UC separations of model compounds enabled by chaotropic/kosmotropic additives

After the definition of the experimental conditions for the generic UC method development procedure, all model compounds described in Fig. 2(a) have been used to create a complete mixture to showcase the separation quality achievable with UC. In addition to the nine nucleobases and nucleosides, a nucleotide (thymidine 3,5-cyclic monophosphate) was added to have a more comprehensive mixture eluting across the entire UC gradient.

In Fig. 3(a), (b), chromatograms for mixture 1 under all combinations of additive and column chemistries are illustrated. As shown, mapping the effect of either chaotropic or kosmotropic salts in unified chromatography has made it possible to quickly develop a method for this multicomponent mixture in a relatively short timeframe. With only six injections, a baseline separation of all 10 nucleosides/nucleobases/nucleotide was achieved on a DEA column with sodium methanesulfonate in the mobile phase (Fig. 3(a)). All other combinations have either failed to ensure either elution of all mixture components (DEA – NaForm; DIOL – NaForm) or baseline separation ($NaClO_4$ on DIOL and DEA; DIOL – NaOMs). The missing elution of the more retained sample on DEA column with sodium formate was somewhat expected, as the same trend is seen in Fig. 2(c) (right graph – green symbols). Similarly, the inability of the chaotropic agent to fully resolve the multicomponent mixture was somewhat forecasted in Fig. 2(c). With the DIOL column on the other hand, full elution is never achieved, thus making this column option the least preferred one. In conclusion, a successful hit has been quickly identified, enabling a rapid method development and optimization for the model mixture with the delivery of a fast separation in under ten minutes.

To better understand the performance of this method development approach for drug development, a second mixture (Fig. 4(a)) was selected and utilized. Mixture 2 is comprised of pharmaceutical "building blocks" (benzotriazoles; carbazoles; azaindoles etc.), compounds which are generally used at different stages of the API synthesis. While mixture 1 was composed of highly retained analytes that required UC-type gradients, mixture 2 is eluted with a more typical SFC gradient (max. 30–40 % of modified blended with $scCO_2$). Therefore, it was selected to evaluate whether chaotropic/kosmotropic salts can provide a different separation profile even with limited amounts of modifier in the mobile phase.

Mixture 2 has been initially screened using the protocol introduced herein, with the generic UC-gradient already employed in the previous case study (Fig. 3). Although full separation of all ten components was not achieved at this point, the initial screening step was nonetheless extremely useful in aiding with the stationary phase selection (Fig. 4(b)). From the scorecard it is evident that, for mixture 2, the DIOL column delivers a promising chromatographic separation under all additives included in the screening step. On the other hand, the DEA stationary phase did not perform at the same level as the DIOL, therefore it was not considered for the subsequent method optimization phase. Due to the relatively low retention of mixture 2 compounds, a switch from the generic UC gradient to a more typical SFC one was made to better highlight potential differences between the chaotropic/kosmotropic salts (Fig. 4(c)). Three elution regions can be identified: for the early eluting molecules (compounds 1–3) no major differences were observed, as they all presented the same elution order as well as baseline separation with either sodium perchlorate, methanesulfonate and formate (Fig. 4(c)). A second elution area, constituted of compounds 4–7, was particularly challenging to fully separate. Here, the choice of the additive (sodium formate) proved to be extremely important in delivering the best separation quality, specifically for the positional isomers 4-methyl-1H-benzotriazole and 5-methyl-1H-benzotriazole (Fig. 4(d)): using the kosmotropic formate anion may have impacted the water structure on the surface of the stationary phase, as well as the solvation sphere of the two analytes, so to improve the overall chromatographic separation achievable. A more in-depth investigation is, however, needed to better understand the observed phenomenon. Sodium formate also provides the best overall separation in this elution area (Fig. 4(d)). Finally, the third elution region, corresponding to the more retained analytes, also sees a distinctive difference between the additives.



(caption on next column)

Fig. 4. (a) Structures of pharmaceutical building blocks used in mixture 2: (1) 1-(methoxymethyl)-1*H*-benzotriazole; (2) *N*-(4-formylpentyl)carbazole; (3) 6-methoxy-7-azaindole; (4) 3-[5-(4-chlorophenyl)furan-2-yl]acrylic acid; (5) 4-methyl-1*H*-benzotriazole; (6) 5-methyl-1*H*-benzotriazole; (7) 1*H*-benzotriazole-1-methanol; (8) benzotriazole; (9) 7-azaindole-carboxyaldehyde; (10) benzotriazole-5-carboxylic acid. (b) scorecard from screening step, highlighting the number of analytes fully separated in each combination of column and additive. (c) Comparison of sodium perchlorate (blue), sodium methanesulfonate (grey) and sodium formate (green) using the final optimized method. All additives were concentrated at 5 mM in 95:5 MeOH:H_2O co-solvent. Column: Torus DIOL. Gradient profile: hold at 2 % for 0.5 min, followed by a first gradient ramp to 7 % in 1.5 min, a second ramp to 20 % in 3.0 min, and a third ramp to 35 % in 3.0 min, hold-up at 65:35 $scCO_2$:B for 0.5 min, then return to initial conditions in 0.1 min. Total run time: 10 min. Flow-rate: 1.50 mL min^{-1}. Column temperature: 45 °C. ABPR: 110 bar. Detection: UV @ 254 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 1.5 μL. d) Zoom-in of chromatograms depicted above (3c), to highlight separation of compounds 4–5–6–7 with different additives.(For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Sodium perchlorate delivers the best separation while also ensuring the elution of all analytes, followed by sodium methanesulfonate. Sodium formate improves the resolution between compounds 8 and 9, however fails to elute compounds 10 (benzotriazole-5-carboxylic acid) with the gradient ramp utilized so far. An optimized gradient has been, therefore, developed to ensure elution of benzotriazole-5-carboxylic acid and the overall separation of the multicomponent mixture (Fig. 5).

3.4. UC vs. RP-HPLC for forced degradation study sample

The final step of this study is to evaluate unified chromatography, and the use of chaotropic/kosmotropic salts in $scCO_2$-based mobile phases, with a real-life example from a compound currently under development in AbbVie's pipeline. RPLC remains the golden standard in the industry, nonetheless alternative chromatographic modalities have been increasingly used to complement it. SFC/UC, thanks to its nature, has constantly demonstrated to deliver orthogonal separation profiles to RPLC, thus making it an excellent alternative for challenging separations.

In Fig. 6 and Table 1, a comparison between unified chromatography and reversed-phase HPLC is highlighted for a pipeline compound (AbbVie-1), which has undergone forced degradation study at high temperature (70 °C), using both a low humidity (10 % relative humidity – RH) and high humidity (75 % RH) environments for ten days. Forced degradation studies are very important in pharmaceutical drug development, as they enable scientists to quickly gather stability data for a given sample, as well as identifying major impurities. The use of extreme conditions, however, can trigger the generation of complex reaction mixtures which can be very challenging to separate. RP-HPLC can deliver a satisfactory separation of the main compound AbbVie-1 from the main impurities with a peak area (PA) higher than 0.1 % for the sample stressed at high temperature (70 °C) and low humidity (10 % RH) (Fig. 6(c) - blue). However, when a high humidity environment was used to stress the selected compound, RP-HPLC did not deliver the same performance seen before. The chromatogram (Fig. 6(c) - gold) indicates the presence of an elevated number of early eluted species (Table 3), demonstrating the relative inability of RP-HPLC to efficiently retain highly polar compounds. Due to the relatively high peak area of the early eluting species, constituting the largest impurity in the analysis (Table 3), it is interesting to gather more data on those analytes, while retaining the same good performance for the more apolar species. Thanks to its ability to simultaneously retain and elute apolar and polar species, UC was considered as a viable alternative to generate a better chromatographic separation than RP-HPLC.

After the rapid screening stage (Fig. 6(a)), the most promising

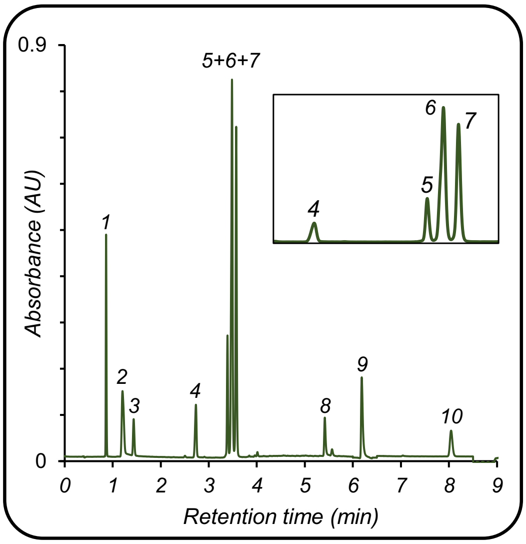


Fig. 5. Optimized separation of mixture 2 using 5 mM sodium formate in the co-solvent (95:5 MeOH:H_2O). Gradient profile: hold at 2 % for 0.5 min, followed by a first gradient ramp to 7 % in 1.5 min, a second ramp to 20 % in 3.0 min, and a third ramp to 45 % in 3.0 min, hold-up at 55:45 $scCO_2$:B for 0.5 min, then return to initial conditions in 0.1 min. Total run time: 10 min. Flow-rate: 1.50 mL min^{-1}. Column temperature: 45 °C. ABPR: 110 bar. Detection: UV @ 254 nm with compensated mode (compensation range: 350–450 nm). Injection volume: 1.5 μL.

conditions have been identified with the combination of sodium perchlorate and the use of a DIOL column. Fig. 6(b) and Table 3 clearly illustrate the advantages of UC over the established RP-HPLC method. The complementarity between the two techniques is highlighted by the shift in retention time of the main compound AbbVie-1 (~16.0 min in RP-HPLC; ~1.0 min in UC). All the impurities generated during the accelerated stability study are more retained than AbbVie-1 using unified chromatography, while in RP-HPLC and with the use of 70 °C and 75 % RH conditions, majority of the impurities elute before the main compound. The higher retentivity ensured by UC translates into the retention of those chemical species eluting at the column void under RP-HPLC conditions (Fig. 6(b) and (c) – gold chromatograms). UC does not only enable a better chromatographic profile of the stressed AbbVie-1 sample, but also provides a better separation overall (Table 3). A higher number of peaks, having a peak area (%) value equal to or higher than 0.1 % is seen with UC. Lower purity of AbbVie-1 is obtained with UC for both stressing conditions, as well as for the largest impurity (% PA) in each sample: this is due to the better separation power generated with the UC method, which can separate species co-eluting with AbbVie-1.

4. Conclusions

In this work, a new strategy for a quick and streamlined UC method development in the drug development environment is introduced. This approach systematically utilizes unorthodox additives, such as sodium perchlorate, sodium methanesulfonate and sodium formate to efficiently exploit chaotropic/kosmotropic effects in UC. Fundamental studies have highlighted the importance of using at least 5 % of water blended with methanol in the co-solvent, and overall illustrated the differences between low (2 %) and high (7 %) percentages of water in the modifier. Using this expanded knowledge, SFC/UC methods have been developed for a series of challenging multicomponent mixtures. The method development procedure here proposed has enabled the rapid selection of the best combination of co-solvent additive and column chemistry, with a full screening requiring only one hour per compound. Chaotropic and kosmotropic salts have been vital in delivering separations for closely-

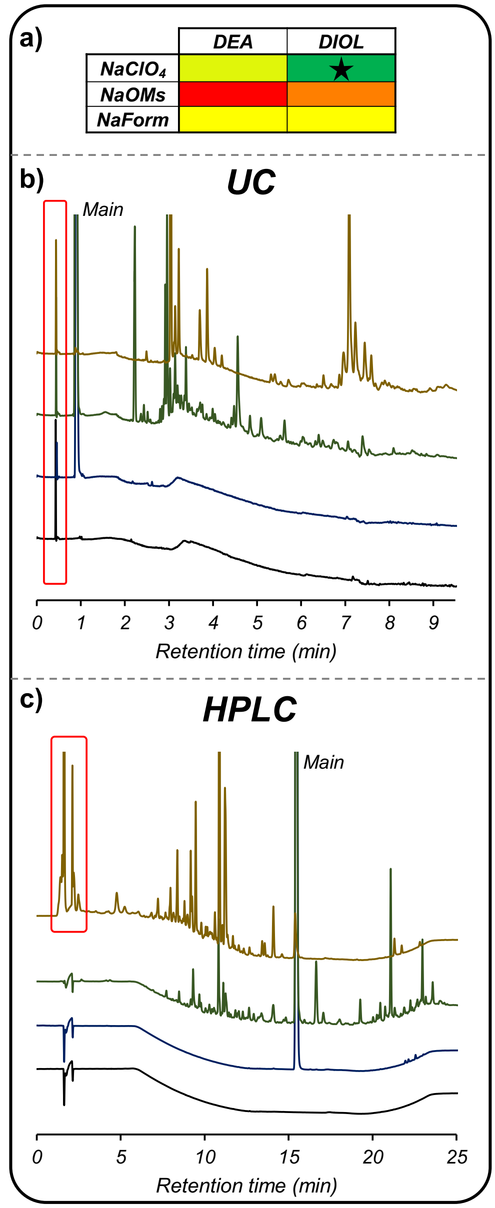


Fig. 6. (a) scorecard from general UC screening, starred box indicates best conditions identified. (b) UC separations for compound AbbVie-1 control sample (blue), 10 days stressed sample at 70 °C – 10 % relative humidity (green) and 10 days stressed sample at 70 °C – 75 % relative humidity (gold). (c) RP-HPLC separations for compound AbbVie-1 control sample (blue), 10 days stressed sample at 70 °C – 10 % relative humidity (green) and 10 days stressed sample at 70 °C – 75 % relative humidity (gold).(For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

related compounds, as well as highly polar (bio)molecules. Finally, a direct comparison of chaotropic-enabled UC vs. RPLC for an experimental compound undergoing forced degradation studies has shown how the new method development strategy here introduced has quickly delivered a better separation compared to the fully-established RP-HPLC technique. This supports the importance, in current method development occurring across all stages of drug development, of exploring alternative and orthogonal techniques to achieve a better separation quality.

Table 3
Comparison UC *vs* RP-HPLC for stressed AbbVie-1 compound.

Technique	Stress condition	No. peaks with %PA > 0.1 %	Purity AbbVie-1 (PA)	Largest impurity (PA)
UC	70 °C – 10 % RH	45	62.68 %	5.43 % (RRT = 2.58)
RP-HPLC	70 °C – 10 % RH	41	68.60 %	5.33 % (RRT = 2.49)
UC	70 °C – 75 % RH	46	0.68 %	22.58 % (RRT = 3.28)
RP-HPLC	70 °C – 75 % RH	42	3.03 %	28.91 % (RRT = 0.21)

Disclosures

All authors are employees of AbbVie and may own AbbVie stock. The design, study conduct, and financial support for this research were provided by AbbVie. AbbVie participated in the interpretation of data, review, and approval of the publication.

CRediT authorship contribution statement

Gioacchino Luca Losacco: Conceptualization, Methodology, Investigation, Visualization, Project administration, Writing – original draft. **Zachary S. Breitbach:** Writing – review & editing. **Paul L. Walsh:** Writing – review & editing. **Leon Van Haandel:** Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

All authors are employees of AbbVie and may own AbbVie stock. The design, study conduct, and financial support for this research were provided by AbbVie. AbbVie participated in the interpretation of data, review, and approval of the publication.

Data availability

Data will be made available on request.

References

[1] Y. Shen, D.-D. Shen, R. Jia, J.-S. Sun, J. Song, H.-M. Liu, New drug approvals for 2022: synthesis and clinical applications, Med. Res. Rev. (2023) 1–40, https://doi.org/10.1002/med.21976.
[2] S. Yuan, D.-S. Wang, H. Liu, S.-N. Zhang, W.-G. Yang, M. Lv, Y.-X. Zhou, S.-Y. Zhang, J. Song, H.-M. Liu, New drug approvals for 2021: synthesis and clinical applications, Eur. J. Med. Chem. (2022) 245, https://doi.org/10.1016/j.ejmech.2022.114898.
[3] J.T. Kuethe, J. Lee, D. Thaisrivongs, N. Yasuda, S.R. Pollack, J. Leone, J. DaSilva, M. Biba, F.-R. Tsay, E.L. Regalado, J. Qi, H. Li, G. Dal Poggetto, R. Cohen, Synthesis of a complex and highly potent PCSK9 inhibitor, Org. Lett. (2023) 25, https://doi.org/10.1021/acs.orglett.3c01635.
[4] E.L. Regalado, I.A. Haidar Ahmad, R. Bennett, V. D'Atri, A.A. Makarov, G. R. Humphrey, I. Mangion, D. Guillarme, The emergence of universal chromatographic methods in the research and development of new drug substances, Acc. Chem. Res. 52 (2019) 1990–2002, https://doi.org/10.1021/acs.accounts.9b00068.
[5] A.A. Dar, P.L. Sangwan, A. Kumar, Chromatography: an important tool for drug discovery, J. Sep. Sci. 43 (2020) 105–119, https://doi.org/10.1002/jssc.201900656.
[6] S. Fekete, M.K. Aebischer, M. Imiołek, T. Graf, R. Ruppert, M. Lauber, V. D'Atri, D. Guillarme, Chromatographic strategies for the analytical characterization of adeno-associated virus vector-based gene therapy products, TrAC Trends Anal. Chem. 164 (2023), 117088, https://doi.org/10.1016/j.trac.2023.117088.
[7] A. Goyon, L. Dai, T. Chen, B. Wei, F. Yang, N. Andersen, R. Kopf, M. Leiss, M. Mølhøj, D. Guillarme, C. Stella, From proof of concept to the routine use of an automated and robust multi-dimensional liquid chromatography mass spectrometry workflow applied for the charge variant characterization of therapeutic antibodies, J. Chromatogr. A 1615 (2020), 460740, https://doi.org/10.1016/j.chroma.2019.460740.
[8] X. Wang, G. Yan, H. Zheng, M. Gao, X. Zhang, Strategy for high-throughput identification of protein complexes by array-based multi-dimensional liquid chromatography-mass spectrometry, J. Chromatogr. A 1652 (2021), 462351, https://doi.org/10.1016/j.chroma.2021.462351.
[9] S.R. Groskreutz, M.M. Swenson, L.B. Secor, D.R. Stoll, Selective comprehensive multi-dimensional separation for resolution enhancement in high performance liquid chromatography. Part I: principles and instrumentation, J. Chromatogr. A 1228 (2012) 31–40, https://doi.org/10.1016/j.chroma.2011.06.035. High-Perform. Columns Their Oper. Speed Effic. Peak Capacity Obtainable High-Perform. Columns Var. Oper. Cond.
[10] S.H. Yang, J. Wang, K. Zhang, Validation of a two-dimensional liquid chromatography method for quality control testing of pharmaceutical materials, J. Chromatogr. A 1492 (2017) 89–97, https://doi.org/10.1016/j.chroma.2017.02.074.
[11] M. Iguiniz, F. Rouvière, E. Corbel, N. Roques, S. Heinisch, Comprehensive two dimensional liquid chromatography as analytical strategy for pharmaceutical analysis, J. Chromatogr. A 1536 (2018) 195–204, https://doi.org/10.1016/j.chroma.2017.08.070. Compr. Two-Dimens. Chromatogr.
[12] G.L. Losacco, R.D. Cohen, J.O. DaSilva, I.A. Haidar Ahmad, E.C. Sherer, I. Mangion, E.L. Regalado, Deuterated modifiers in sub/supercritical fluid chromatography for streamlined NMR structure elucidation, Anal. Chem. 94 (2022) 12176–12184, https://doi.org/10.1021/acs.analchem.2c02623.
[13] A. Grand-Guillaume Perrenoud, J.-L. Veuthey, D. Guillarme, Comparison of ultra-high performance supercritical fluid chromatography and ultra-high performance liquid chromatography for the analysis of pharmaceutical compounds, J. Chromatogr. A 1266 (2012) 158–167, https://doi.org/10.1016/j.chroma.2012.10.005.
[14] G.L. Losacco, J.-L. Veuthey, D. Guillarme, Metamorphosis of supercritical fluid chromatography: a viable tool for the analysis of polar compounds? TrAC Trends Anal. Chem. 141 (2021), 116304 https://doi.org/10.1016/j.trac.2021.116304.
[15] M.A. Patel, F. Riley, M. Ashraf-Khorassani, L.T. Taylor, Supercritical fluid chromatographic resolution of water soluble isomeric carboxyl/amine terminated peptides facilitated via mobile phase water and ion pair formation, J. Chromatogr. A 1233 (2012) 85–90, https://doi.org/10.1016/j.chroma.2012.02.024.
[16] M. Ashraf-Khorassani, L.T. Taylor, E. Seest, Screening strategies for achiral supercritical fluid chromatography employing hydrophilic interaction liquid chromatography-like parameters, J. Chromatogr. A 1229 (2012) 237–248, https://doi.org/10.1016/j.chroma.2011.12.089.
[17] A. Grand-Guillaume Perrenoud, J.-L. Veuthey, D. Guillarme, Coupling state-of-the-art supercritical fluid chromatography and mass spectrometry: from hyphenation interface optimization to high-sensitivity analysis of pharmaceutical compounds, J. Chromatogr. A 1339 (2014) 174–184, https://doi.org/10.1016/j.chroma.2014.03.006.
[18] C. West, Current trends in supercritical fluid chromatography, Anal. Bioanal. Chem. 410 (2018) 6441–6457, https://doi.org/10.1007/s00216-018-1267-4.
[19] G.L. Losacco, R. Bennett, I.A.H. Ahmad, R.C. Barrientos, J.O. DaSilva, Y. Dong, A. W. Schuppe, Z. Wang, S. Aiken, I. Mangion, V.K. Aggarwal, E.L. Regalado, Dual-gradient unified chromatography: a new paradigm for versatility in simultaneous multicomponent analysis, Angew. Chem. Int. Ed. 61 (2022), e202208854, https://doi.org/10.1002/anie.202208854.
[20] J. Molineau, M. Hideux, P. Hennig, S. Bertin, F. Mauge, E. Lesellier, C. West, Analysis of short-chain bioactive peptides by unified chromatography-electrospray ionization mass spectrometry. Part II. Comparison to reversed-phase ultra-high performance liquid chromatography, J. Chromatogr. A 1663 (2022), 462771, https://doi.org/10.1016/j.chroma.2021.462771.
[21] G.L. Losacco, J.O. DaSilva, J. Liu, E.L. Regalado, J.-L. Veuthey, D. Guillarme, Expanding the range of sub/supercritical fluid chromatography: advantageous use of methanesulfonic acid in water-rich modifiers for peptide analysis, J. Chromatogr. A 1642 (2021), 462048, https://doi.org/10.1016/j.chroma.2021.462048.
[22] R. Bennett, M. Biba, J. Liu, I.A. Haidar Ahmad, M.B. Hicks, E.L. Regalado, Enhanced fluidity liquid chromatography: a guide to scaling up from analytical to preparative separations, J. Chromatogr. A 1595 (2019) 190–198, https://doi.org/10.1016/j.chroma.2019.02.017.
[23] M. Ashraf-Khorassani, L.T. Taylor, Subcritical fluid chromatography of water soluble nucleobases on various polar stationary phases facilitated with alcohol-modified CO_2 and water as the polar additive, J. Sep. Sci. 33 (2010) 1682–1691, https://doi.org/10.1002/jssc.201000047.
[24] M.A. Patel, M.A. Hardink, L. Wrisely, F.W. Riley, C.J. Hudalla, M. Ashraf-Khorassani, L.T. Taylor, Evolution of strategies to achieve baseline separation of ten anionic, water-soluble sulfated estrogens via achiral packed column supercritical fluid chromatography, J. Chromatogr. A 1370 (2014) 240–245, https://doi.org/10.1016/j.chroma.2014.10.021.
[25] G.L. Losacco, O. Ismail, J. Pezzatti, V. González-Ruiz, J. Boccard, S. Rudaz, J.-L. Veuthey, D. Guillarme, Applicability of supercritical fluid chromatography–mass spectrometry to metabolomics. II—Assessment of a comprehensive library of metabolites and evaluation of biological matrices, J. Chromatogr. A 1620 (2020), 461021, https://doi.org/10.1016/j.chroma.2020.461021.
[26] L. Si-Hung, T. Bamba, Current state and future perspectives of supercritical fluid chromatography, TrAC Trends Anal. Chem. 149 (2022), 116550, https://doi.org/10.1016/j.trac.2022.116550.
[27] Y. Marcus, Effect of Ions on the Structure of Water: structure Making and Breaking, Chem. Rev. 109 (2009) 1346–1370, https://doi.org/10.1021/cr8003828.

[28] L. Pan, R. LoBrutto, Y.V. Kazakevich, R. Thompson, Influence of inorganic mobile phase additives on the retention, efficiency and peak symmetry of protonated basic compounds in reversed-phase liquid chromatography, J. Chromatogr. A 1049 (2004) 63–73, https://doi.org/10.1016/j.chroma.2004.07.019.
[29] I.A. Haidar Ahmad, W. Chen, H.M. Halsey, A. Klapars, J. Limanto, G.F. Pirrone, T. Nowak, R. Bennett, R. Hartman, A.A. Makarov, I. Mangion, E.L. Regalado, Multi-column ultra-high performance liquid chromatography screening with chaotropic agents and computer-assisted separation modeling enables process development of new drug substances, Analyst 144 (2019) 2872–2880, https://doi.org/10.1039/C8AN02499E.
[30] Y.V. Kazakevich, R. LoBrutto, R. Vivilecchia, Reversed-phase high-performance liquid chromatography behavior of chaotropic counteranions, J. Chromatogr. A 1064 (2005) 9–18, https://doi.org/10.1016/j.chroma.2004.11.104.
[31] R.C. Barrientos, G.L. Losacco, M. Azizi, H. Wang, A.N. Nguyen, V. Shchurik, A. Singh, D. Richardson, I. Mangion, D. Guillarme, E.L. Regalado, I.A. Haidar Ahmad, Automated hydrophobic interaction chromatography screening combined with in silico optimization as a framework for nondenaturing analysis and purification of biopharmaceuticals, Anal. Chem. 94 (2022) 17131–17141, https://doi.org/10.1021/acs.analchem.2c03453.
[32] C.T. Tomaz, Chapter 7—Hydrophobic interaction chromatography*, in: S. Fanali, P.R. Haddad, C.F. Poole, M.-L. Riekkola (Eds.), Liq. Chromatogr, second ed., Elsevier, 2017, pp. 171–190, https://doi.org/10.1016/B978-0-12-805393-5.00007-5.
[33] M. Rodriguez-Aller, D. Guillarme, A. Beck, S. Fekete, Practical method development for the separation of monoclonal antibodies and antibody-drug-conjugate species in hydrophobic interaction chromatography, part 1: optimization of the mobile phase, J. Pharm. Biomed. Anal. 118 (2016) 393–403, https://doi.org/10.1016/j.jpba.2015.11.011.
[34] J. Liu, A.A. Makarov, R. Bennett, I.A. Haidar Ahmad, J. DaSilva, M. Reibarkh, I. Mangion, B.F. Mann, E.L. Regalado, Chaotropic effects in sub/supercritical fluid chromatography via ammonium hydroxide in water-rich modifiers: enabling separation of peptides and highly polar pharmaceuticals at the preparative scale, Anal. Chem. 91 (2019) 13907–13915, https://doi.org/10.1021/acs.analchem.9b03408.
[35] A.M. Hyde, S.L. Zultanski, J.H. Waldman, Y.-L. Zhong, M. Shevlin, F. Peng, General principles and strategies for salting-out informed by the Hofmeister series, Org. Process Res. Dev. 21 (2017) 1355–1370, https://doi.org/10.1021/acs.oprd.7b00197.
[36] K.P. Gregory, G.R. Elliott, H. Robertson, A. Kumar, E.J. Wanless, G.B. Webber, V.S. J. Craig, G.G. Andersson, A.J. Page, Understanding specific ion effects and the Hofmeister series, Phys. Chem. Chem. Phys. 24 (2022) 12682–12718, https://doi.org/10.1039/D2CP00847E.
[37] B. Kang, H. Tang, Z. Zhao, S. Song, Hofmeister series: insights of ion specificity from amphiphilic assembly and interface property, ACS Omega 5 (2020) 6229–6239, https://doi.org/10.1021/acsomega.0c00237.
[38] G.L. Losacco, E. Marconetto, R. Nicoli, T. Kuuranne, J. Boccard, S. Rudaz, J.-L. Veuthey, D. Guillarme, Supercritical fluid chromatography–mass spectrometry in routine anti-doping analyses: estimation of retention time variability under reproducible conditions, J. Chromatogr. A 1616 (2020), 460780, https://doi.org/10.1016/j.chroma.2019.460780.
[39] A. Raimbault, M. Dorebska, C. West, A chiral unified chromatography–mass spectrometry method to analyze free amino acids, Anal. Bioanal. Chem. 411 (2019) 4909–4917, https://doi.org/10.1007/s00216-019-01783-5.
[40] L. Si-Hung, Y. Izumi, T. Bamba, First proof-of-concept of UC/HILIC for extending the versatility of the current art of supercritical fluid separation, Anal. Chim. Acta. 1240 (2023), 340741, https://doi.org/10.1016/j.aca.2022.340741.